CSC 3204: AUTOMATA THEORY

Lecture X:
Finite Automata with Output

Recap

- Motivation for FA => to design a mathematical model for a computer:
  - Input string => represents the program and input data
  - Reading the letters from the string => executing instructions (it changes the state of the machine i.e. Memory contents, control section of processor etc..)
  - What about output?
- If our FA's and TG's represent actual physical machines, then we should also consider FA with output capabilities, not just as language acceptors or recognizers.

Modeling output

- We consider the output as part of the total state of the machine.
- This could mean two different things:
1. entering a specific computer state means change memory a certain way and print a specific character (reflects what we are now printing), or
2. a state includes both the present condition of memory plus the total output thus far (reflects what we have printed in total)
- **Key question:** Do machines based on these two different models have equal power or are there some tasks that one can do that the other cannot?

Modeling output

- For second option: If printing of output is to be done at the end of the program run ➔ have an instruction that dumps a buffer – what issues arise?
  - Imposes a limit on the maximum number of characters that the program can print ➔ size of the buffer.
  - However, theoretically we should be able to have outputs of any finite length. e.g. to allow printing out a copy of the input string which could be arbitrarily long.
- Abstract models of physical machines (FA's and TG's) have to address these questions if they are to represent actual physical machines.

Topics

- Moore Machines
- Mealy Machines
- An application

A bit of History...

- Moore Machines and Mealy Machines are two models for FA's with output.
- Mealy Machines were created by G. H. Mealy in 1955 while Moore Machines were created (independently) by E. F. Moore in 1956.
- The original purpose of the inventors was to design a mathematical model for sequential circuits (which are only one component of a computer)

Moore Machines - Definition

- A Moore Machine consists of:
  - A finite set of states $q_0, q_1, q_2\ldots$ where q_0 is designated the start state
  - An alphabet of input letters $\Sigma = \{a, b, c\ldots\}$
  - An alphabet of output characters $\Gamma = \{x, y, z\ldots\}$
  - A transition table that shows for *each* state and *each* input letter what state is reached next.
  - An output table that shows what character from Γ is printed by each state that is entered

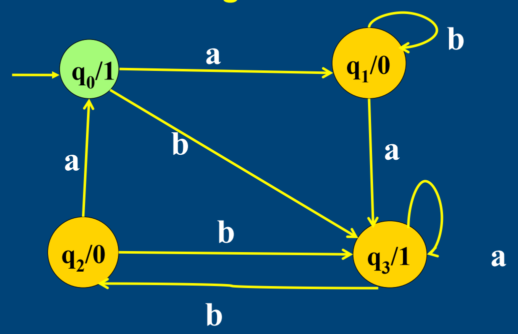
An example of a Moore Machine

- Input alphabet $\Sigma = \{a, b\}$
- Output alphabet $\Gamma = \{0,1\}$
- Names of states: q_0, q_1, q_2, q_3, (q_0 = start state)

An example of a Moore Machine (cont.)

Old state	*New state*		*Output*
	After input a	*After input b*	*(Printed in old state)*
- q_0	q_1	q_3	1
q_1	q_3	q_1	0
q_2	q_0	q_3	0
q_3	q_3	q_2	1

Transition diagram of Moore Machine



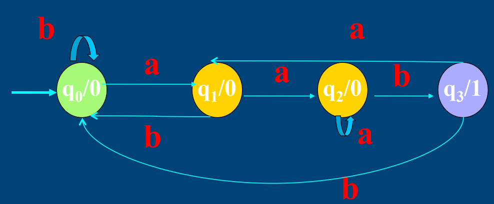
Trace the operation of this machine on the input string *abab*.

A Moore string counter for 'aab'…

Define a moore machine that counts exactly how many times the substring aab occurs in a long input string belonging to a language defined over the alphabet $\Sigma = \{a, b\}$

Trace the operation of this machine on the string aaababbaabb.

A Moore string counter for 'aab'…



This moore machine will count exactly how many times the substring aab occurs in a long input string. Trace the operation of this machine on the string aaababbaabb.

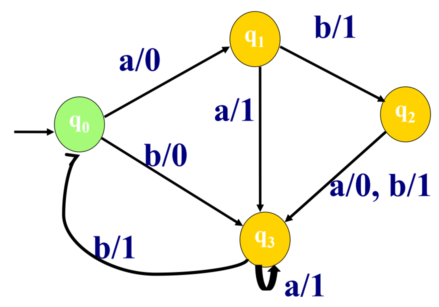
Use bit collection methods to determine how many 1's are in the output string!

Mealy Machines - Definition

A Mealy machine consists of:

- A finite set of states q_0, q_1, q_2… where q_0 is designated the start state
- An alphabet of input letters $\Sigma = \{a, b, c…\}$
- An alphabet of output characters $\Gamma = \{x, y, z…\}$
- A transition diagram where each edge is labelled with a compound symbol *i/o* where *i* is an input letter and *o* is an output character. Every state must have one edge for each possible input letter.

An example Mealy machine

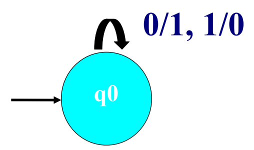


Trace the operation of this machine on the string aaabb

Some useful Mealy machines

- 1's complement (of a binary number)
- A binary increment machine

1's complement Mealy machine



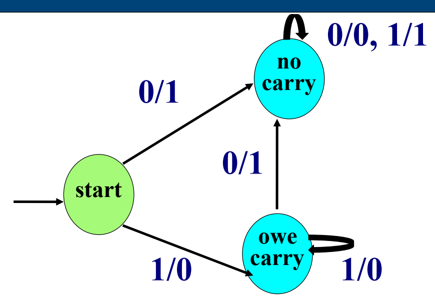
What is the output of this Mealy machine for the input string: 001010 ?

What is the equivalent 1's complement Moore machine?

Some useful Mealy machines

- 1's complement (of a binary number)
- A binary increment machine

Increment Mealy machine



Trace the operation of this Mealy machine on the binary representation of decimal 11.

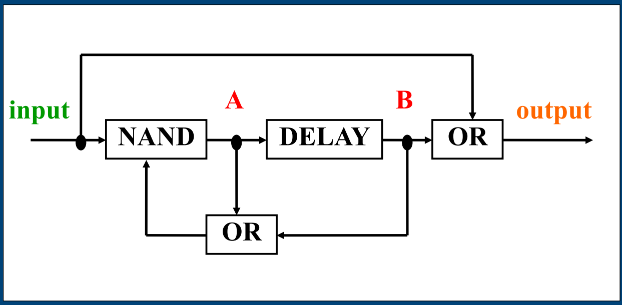
What of decimal 15?

How can we build a subtracting machine from the two example Mealys?

An Application – using a Mealy machine to represent a sequential circuit

- Refresher on Logic gates:
  - NAND => not and
  - DELAY => D-flip flop delays the transmission of the signal along the wire by one step (clock pulse)

Example sequential circuit



State representation of circuit

	A	B
q_0	0	0
q_1	0	1
q_2	1	0
q_3	1	1

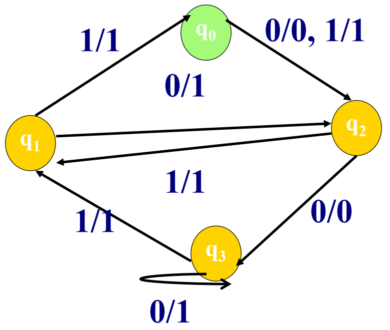
Circuit Logic

- The state of the circuit changes whenever a 1 or 0 is input, based on the logic of its gates as follows
  new B = old A
  new A = (input) **NAND** (old A **OR** old B)
  output = (input) OR (old B)
- At various discrete pulses of a time clock input is received, the state changes and output is generated.

Transition table

Old state	After input 0		After input 1	
	new state	output	new state	output
q_0	q_2	0	q_2	1
q_1	q_2	1	q_0	1
q_2	q_3	0	q_1	1
q_3	q_3	1	q_1	1

Transition Diagram



Summary

- Finite automata with output
- Moore machines output at state
- Mealy machines output at edges
- Moore and Mealy machines are equivalent in expressive power
- Moore and Mealy machines have useful applications such as representation of sequential circuits

Another Application of FA with output – A bilingual dictionary

- Exploits the idea of relations (set theory)
- e.g. {“cat”,”chat”}, {“girl”,”fille”}, {“table”,”table”}, {“house”,”maison”}....
- Such an FA would both accept L_1 and output the corresponding $L_2$where $\{L_1,L_2\}$.
- This machine is both a language acceptor (recognizer) but has output capabilities!

Exercises

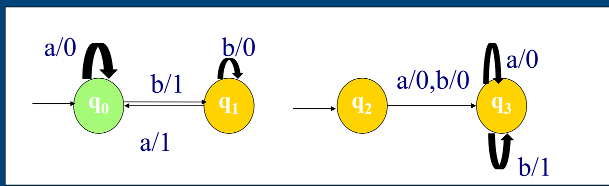
- Each of the following is a moore machine with input alphabet Σ = {a, b} and output alphabet Γ = {0, 1}. Draw the corresponding machines given the following transition and output tables:

	a	b	Output
q0	q1	q2	1
q1	q1	q1	0
q2	q1	q0	1

	a	b	Output
q0	q1	q2	0
q1	q2	q3	0
q2	q3	q4	1
q3	q4	q4	0
q4	q0	q0	0

Exercises

Construct the transition table for each of the two Mealy machines shown below:



Exercises

- Draw a Mealy machine that checks if a very long bit string contains the pattern defined by the following regular expression: *(0 (0 + 1) 1)*. The machine should output a *1* each time the pattern is encountered.
- Describe the string processing task being performed by this Moore machine